

# Dominic Bair

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🐙 github.com/DothWalrus

## Research Interests

My research lies at the intersection of geometric group theory and probability, with emphasis on boundary measures, symbolic dynamics, random metrics, and relatively hyperbolic groups. I study how finite measures and random processes encode the asymptotic geometry of groups, including through Patterson–Sullivan theory and first-passage percolation.

## Education

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| 2022 – Present | <b>Ph.D. candidate, Mathematics</b> , University of Tennessee, Knoxville.<br>Advisors: Yulan Qing and Jan Rosiński.               |
| 2020 – 2022    | <b>M.S. Mathematics</b> , Montana State University.<br>Thesis: <i>DVM: A deep learning algorithm for minimizing functionals</i> . |
| 2016 – 2020    | <b>B.S. Mathematics; Minor in Physics</b> , Montana State University.                                                             |

## Publications and Preprints

- [1] **D. Bair**, S. Jana, and Y. Qing, *First-passage percolation, non-positive curvature, and radial maps*, Preprint, 2025. arXiv: 2512.05720 [math.MG].
- [2] C. Potts, J. Schearer, T. A. Sebrell, **D. Bair**, B. Ayler, J. Love, J. Dankoff, P. R. Harris, D. Zosso, and D. Bimczok, “MNPmApp: An image analysis tool to quantify mononuclear phagocyte distribution in mucosal tissues,” *Cytometry Part A*, vol. 101, no. 12, pp. 1012–1026, 2022.  DOI: 10.1002/cyto.a.24657.

## Academic Appointments

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|----------------|----------------------------------------------------------------------------------------------------------|
| 2022 – Present | <b>Graduate Teaching Assistant</b> , University of Tennessee, Knoxville.                                 |
| 2020 – 2022    | <b>Graduate Teaching Assistant</b> , Montana State University.                                           |
| 2019 – 2020    | <b>Undergraduate Research Assistant</b> , Department of Mathematical Sciences, Montana State University. |
| 2018 – 2019    | <b>Undergraduate Research Assistant</b> , Department of Physics, Montana State University.               |

## Teaching Experience

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| UTK | Statistical Reasoning; Basic Calculus; Mathematical Reasoning; College Algebra. |
| MSU | Calculus I; Precalculus; Survey of Calculus; College Algebra.                   |

## Selected Talks

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- 2026 *Ordered frequency in relatively hyperbolic groups*, UTK Topology Seminar.
- 2026 *First-passage percolation, non-positive curvature, and radial maps*, YGGT XIV.
- 2026 *Graphons and random graphs*, UTK Graduate Student Seminar.
- 2025 *Introduction to first-passage percolation and subadditive processes*, UTK Probability Seminar.
- 2025 *First-passage percolation on hyperbolic graphs: bi-infinite geodesics, radial maps, and infinitely many thick triangles*, UTK Topology Seminar.
- 2020 *Spatial statistics in histology images*, SIAM Northern States Section Student Chapters Conference.
- 2020 *Spatial statistics in histology images*, Data Science and Image Analysis Conference of the Pacific Northwest.

## Additional Projects

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- 2026 **Quanto Options**, Erdős Institute Quantitative Finance Bootcamp.  
Joint work with Kevin Dao, Chloe Ireland, Ethan Katz, and Rahul Pradeep.  
Developed stochastic and numerical models for pricing and hedging quanto options, whose underlying assets and payouts are denominated in different currencies.
- 2022 **DVM: A deep learning algorithm for minimizing functionals**, M.S. thesis, Montana State University.  
Developed a neural-network method for approximating minimizers of variational problems, with applications to partial differential equations.

## Honors and Awards

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- 2026 **Graduate Student Academic Achievement Award**, University of Tennessee, Knoxville.
- 2020 **NSF and AWM Travel Grant**, Data Science and Image Analysis Conference of the Pacific Northwest.
- 2019 – 2020 **INBRE Grant**, support for research in spatial statistics in histology images.
- 2016 – 2020 **Premier Scholarship**, Montana State University.
- 2019 **Superior Achievement in Mathematics**, Pi Mu Epsilon.

## Service and Leadership

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- 2026 – Present **Senator**, UTK Graduate Student Senate.
- 2026 – Present **Leader**, UTK Mathematics Graduate Student Seminar.

## Technical Skills

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- Programming Python, MATLAB, Julia, Git, and  $\text{\LaTeX}$ .
- Methods Numerical simulation, statistical computing, scientific software development, and mathematical modeling.